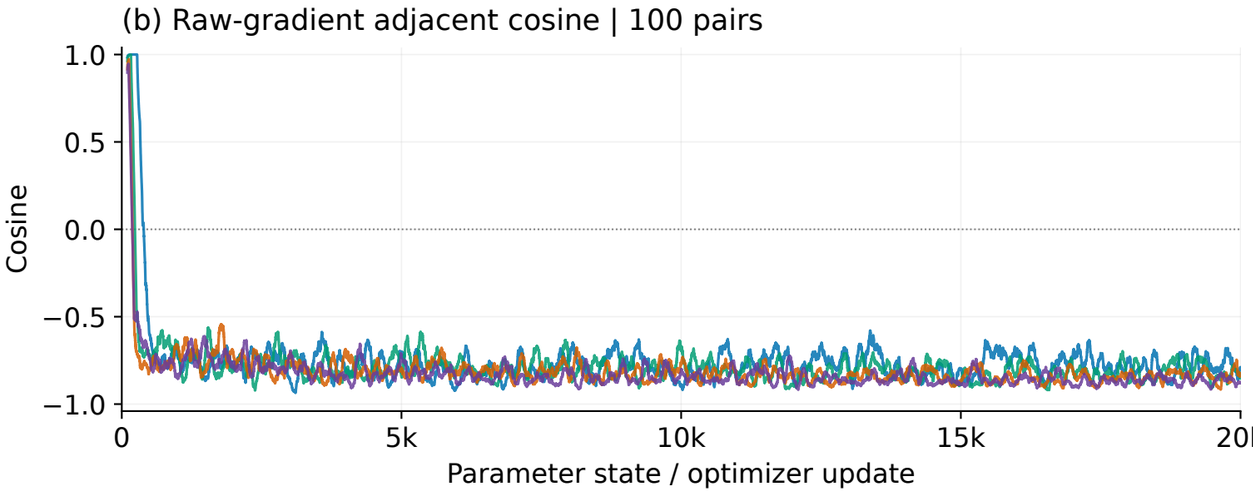
CIFAR-10 | Adam geometry | four learning rates | 20,000 updates

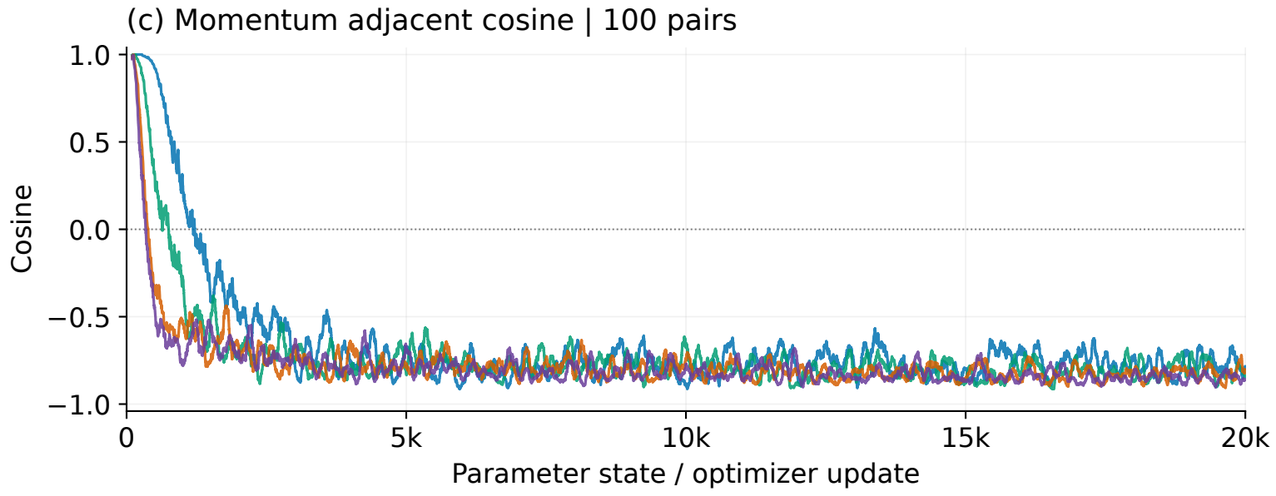
(a) Unsmoothed full-batch training loss



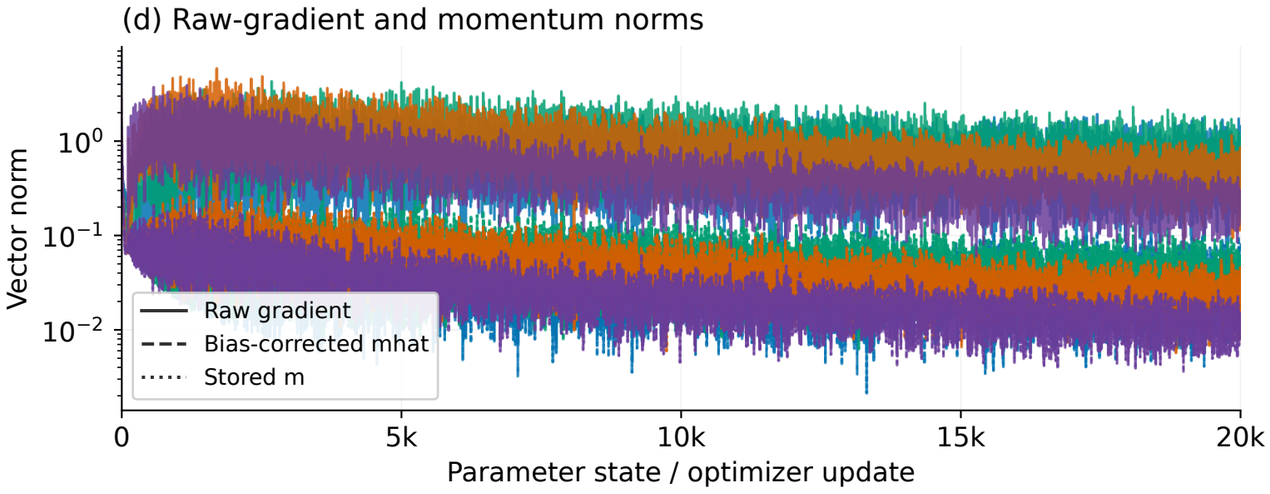
(b) Raw-gradient adjacent cosine | 100 pairs



(c) Momentum adjacent cosine | 100 pairs



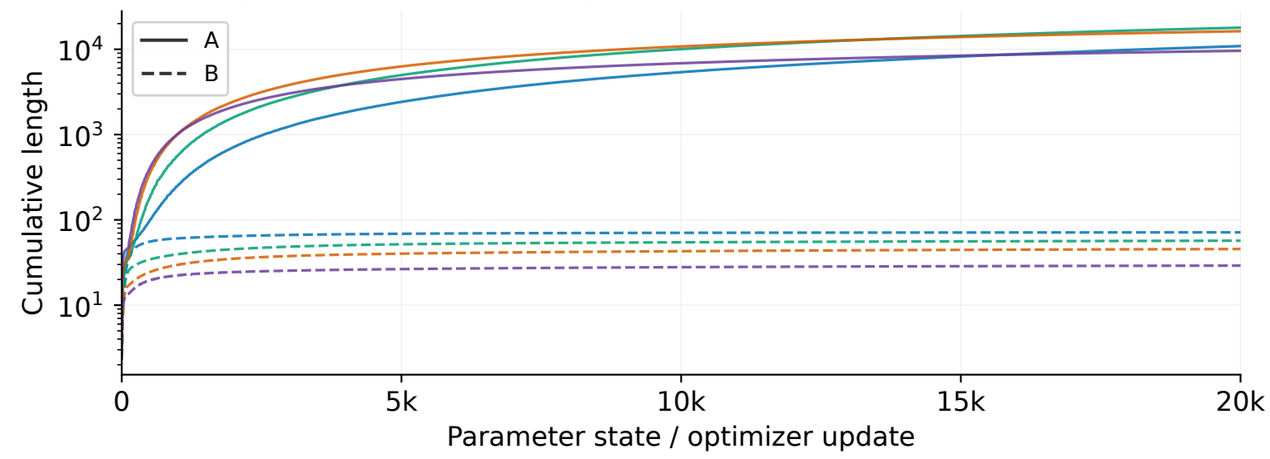
(d) Raw-gradient and momentum norms



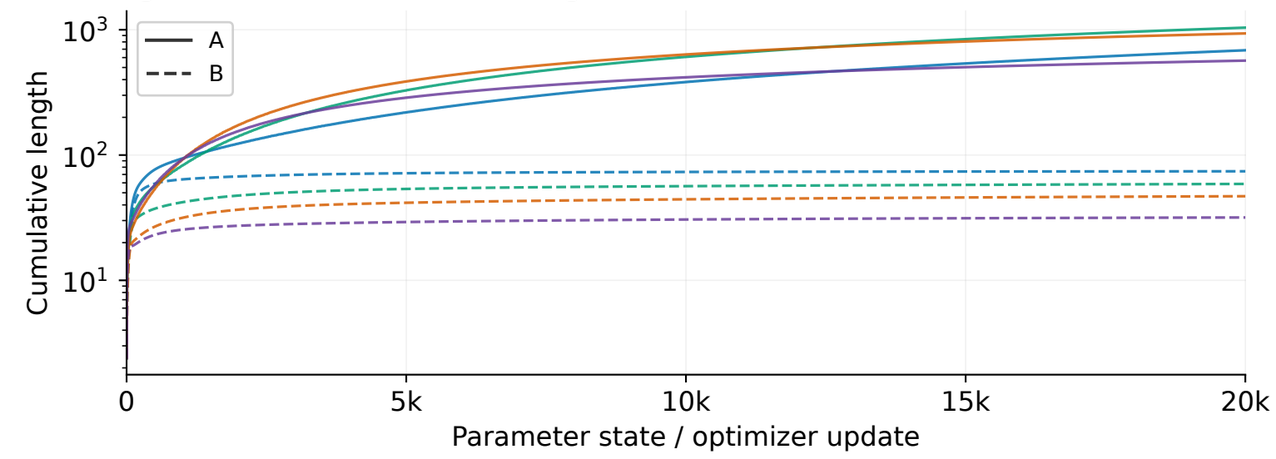
(e) Adaptive direction and actual update norms



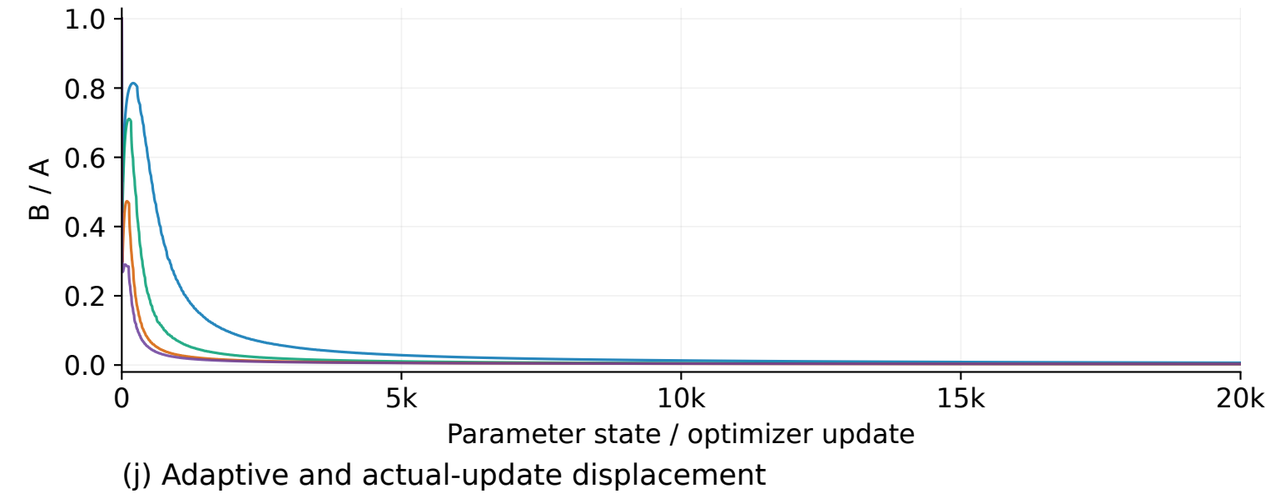
(f) Raw-gradient cumulative lengths



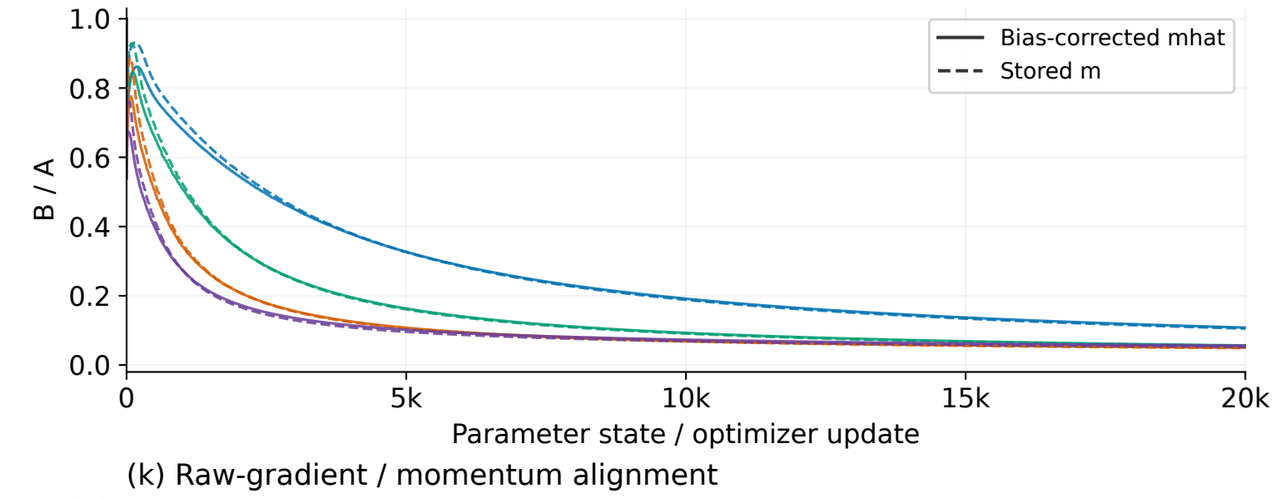
(g) Momentum cumulative lengths



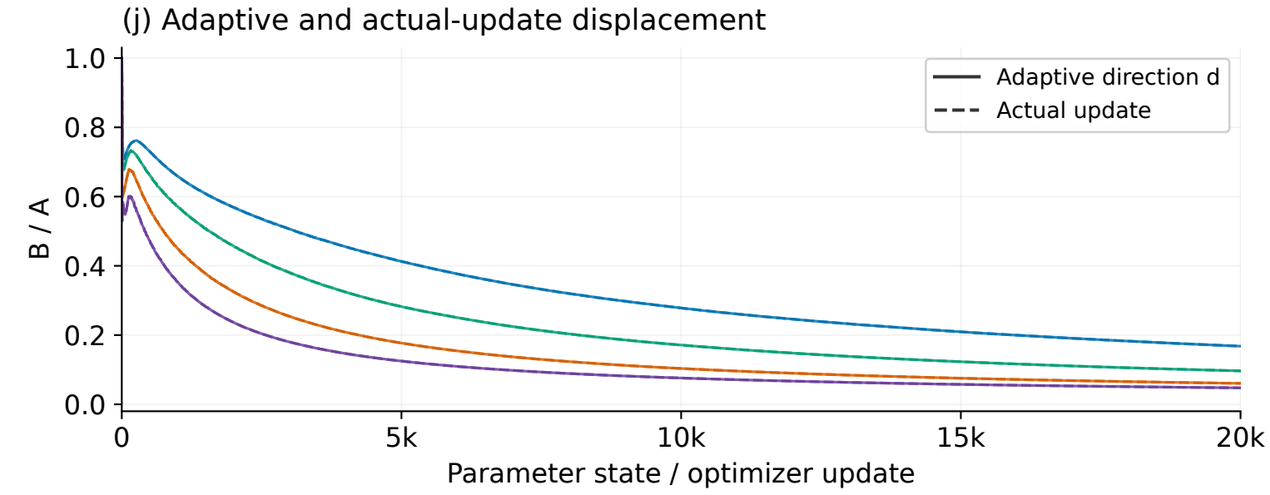
(h) Raw-gradient cumulative displacement



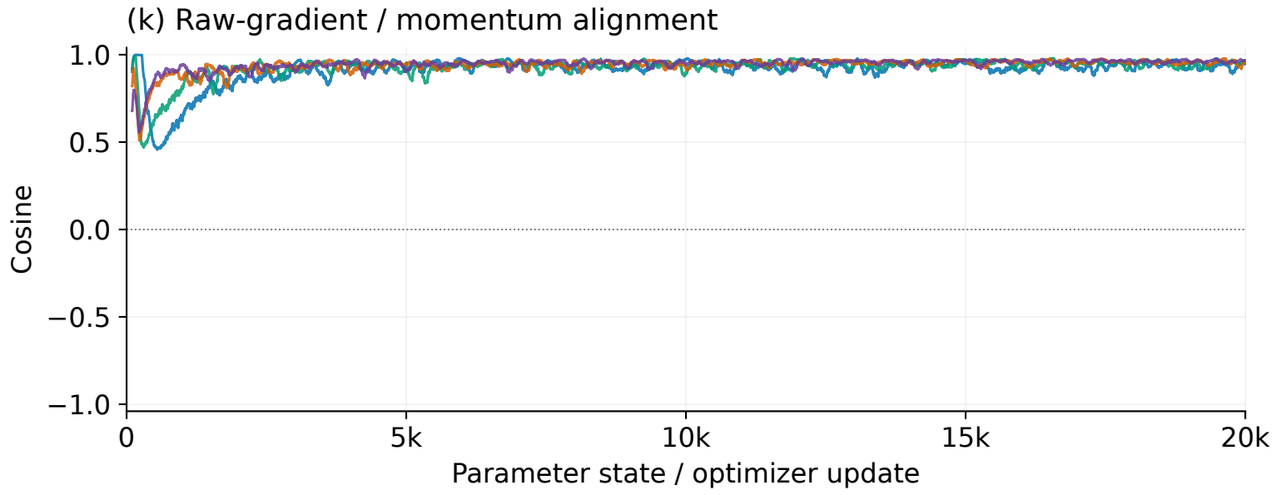
(i) Momentum cumulative displacement



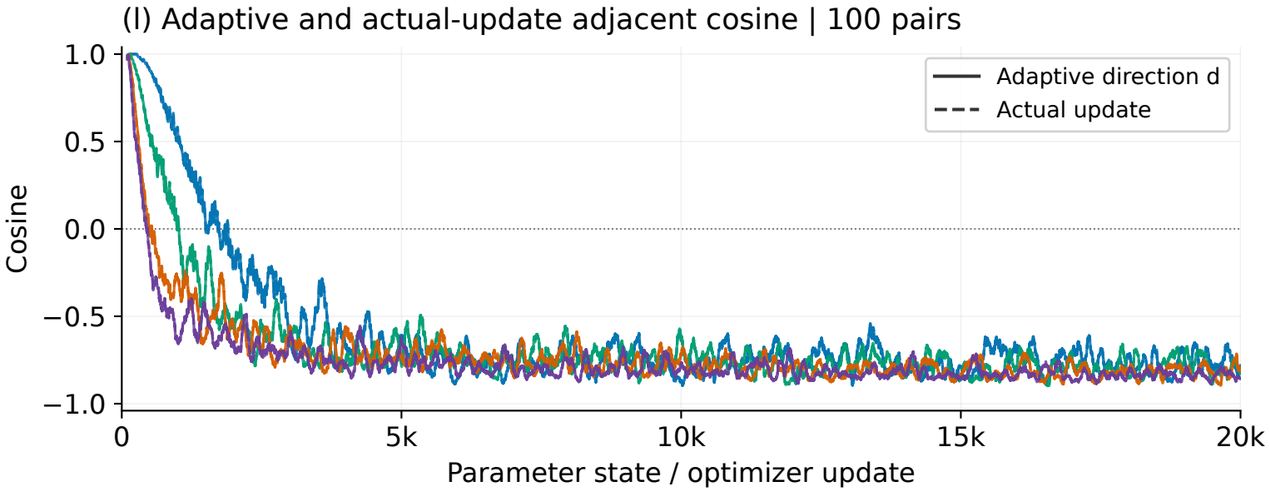
(j) Adaptive and actual-update displacement



(k) Raw-gradient / momentum alignment



(l) Adaptive and actual-update adjacent cosine | 100 pairs



(m) Vector norm ratios



Color identifies LR. A=sum of vector norms; B=norm of the vector sum; ratios use the full prefix.
Momentum means bias-corrected mhat unless explicitly labeled stored m. Actual updates include FP32 rounding. All cosine means use 100 complete pairs.